IRB Administration Office
87 E. Canfield, Second Floor
Detroit, MI 48201
www.research.wayne.edu/irb

Notice of IRB Administrative Determination: Non-Human Participant Research (HPR)

To: Gregory Schwing
From: Helen Craigie D.Be
Date: September 29, 2026
Re: Open Spine Consortium: Secondary Analysis of a De-Identified Radiograph Dataset (VinDr-SpineXR) Obtained Under the PhysioNet Restricted Health Data Use Agreement -- supplement to HPR Determination 2026-269

WSU IRB HPR Number: **2026 330**

Materials in the above referenced proposal were received by the Wayne State University Institutional Review Board Administration Office via Human Participant Research (HPR) Determination Tool on 09/29/26

This project does not constitute human participant research according to the definition codified in the Common Rule at 45 CFR 46 and FDA regulations. This means that IRB review and oversight is not required for this project. We recommend you keep this memo for your records.

The project also does not involve individuals who would receive a test article (drug or device) as participants and therefore the FDA regulations do not apply. Thus, this project does not require review or approval by the Wayne State University Institutional Review Board.

Please be aware that projects involving identifiable Personal Health Information (PHI) which the IRB has determined does not meet the definition of human participant research must still adhere to HIPAA regulations, and may require authorization from the covered entity's privacy officer*.

Projects must also obtain support from any unaffiliated institution(s) involved.

Please contact applicable affiliates: (DMC, Karmanos Cancer Institute, J.D. Dingell Veterans Administration Veterans Medical Center) to obtain authorization to conduct your project.

Please note that changes to the study plan may impact this determination as to whether the project constitutes Human Participant Research. Please contact the IRB Administration office if there are changes to the study plan that may affect this determination.

Helen Craigie
Digitally signed by Helen Craigie
Date: 2026.09.29 14:54:00 -04'00'

Wayne State University IRB Administration Office

Human Participant Research Determination Tool

According to federal regulations from the Department of Health and Human Services (DHHS) and the Food and Drug Administration (FDA), all human subject/participant research must be reviewed and approved by the Institutional Review Board (IRB). This determination tool is designed to determine whether an activity is considered human participant research under either regulation. The IRB Administration Office can provide assistance with making this determination.

Human Participant Research (HPR) Guidance:

Before completing this determination tool, we recommend reviewing the guidance available to help understand how the IRB applies the regulatory definitions to the HPR Determination reviews:

- Does My Study Need IRB Review: Part 1- On- Demand video
- Does My Study Need IRB Review: Part 2- On- Demand video
- HPR Guidance Document

If you determine that the project does not require IRB review:

If by the use of this tool, you have determined that the project does not require IRB review, you do not need to submit this form to the IRB office unless you need documentation from the IRB to concur with your self-determination. Add the project title and name of the person completing the project, their title and the date the tool was completed to the first page, and retain this tool in your files to document this determination.

If there are any modifications to your project that could change this determination, please complete this tool again. Submit the appropriate application to the IRB if changes to your project result in the determination that human participant research is involved according to this tool, which is based on the federal regulations.

If you determine that IRB review is required:

If you have determined that IRB review is required, IRB approval must be obtained **before** conducting any human participant research activities. Visit the WSU IRB website for additional information and the forms required for a new submission: www.research.wayne.edu/irb . E-mail irbquestions@wayne.edu with any questions that come up along the way.

If assistance is needed, or if written documentation from the IRB office is required:

Complete the **entire** form and email it to IRBQuestions@wayne.edu. Include any relevant supporting documents (e.g., grant, protocol, data collection tools). Please do not submit handwritten documents to the IRB office.

IRB Administration Office
87 E. Canfield, Second Floor
Detroit, MI 48201
Telephone# (313) 577-1628
www.research.wayne.edu/irb

IRB Determination Number (IRB Use Only)

2026 330

Section A: Project Staff and Location Information:

Complete all sections of this form if you will be requesting assistance from the IRB Administration Office in making your Human Participation Research determination. Otherwise continue on to Section D to begin the determination tool.

Project Title:	Open Spine Consortium: Secondary Analysis of a De-Identified Radiograph Dataset (VinDr-SpineXR) Obtained Under the PhysioNet Restricted Health			
Name of person conducting the project:	Gregory Schwing	Title:	Resident Physician, General Surgery, Detroit Medical Center / Wayne State University School of Medicine	Date: 09/25/2026
Email Address:	go2432@wayne.edu	Phone:	504-994-0403	
Status: **Select all that apply**	☐ Wayne State Faculty ☐ WSU Graduate Student ☐ WSU Undergraduate Student ☑ Resident/Fellow/Trainee ☐ Other:	☐ DMC Staff ☐ Karmanos Staff ☐ J. D. Dingell VAMC Staff		
Division or College:	School of Medicine	Department:	Surgery	
Campus Address:	5057 Woodward Ave, Detroit, MI 48202			
Faculty Sponsor/ Supervisor for this Project:	Name: Loren Schwiebert ☐ I do not have a Faculty Sponsor/Supervisor	Title: Phone:	Professor, Department of Computer Science, Wayne State University	
Form completed by:	Gregory Schwing	E-mail:	go2432@wayne.edu	
Check ALL that apply:	☐ Behavioral, social, education, non-medical study ☑ Medical study			

Useful Tips:

1. Carefully read through the definitions provided within the determination tool prior to answering the questions. This will help to ensure an accurate determination is made.
2. If your research involves the use of de-identified data or bio-specimens, include a letter of support from the institution/department providing the data/bio-specimens that confirms that the data/bio-specimens provided will be stripped of all identifiable information before you receive it.
3. If your project requires you to **access**, or **use** identifiable data to complete your data collection, you should check yes to #2.2 in Section D even if the data you are retaining/collecting for analysis will be de-identified.
    a. For example, accessing a patient's medical record to collect data for a chart review gives you access to individually identifiable data even though you have no plans to retain/collect any identifiable data. This is considered a human participant as described by #2.2 under section D of this tool.
4. Review the guidance available on the cover page

Section B: Location and Study Data Details:

1.	Describe the location(s) where activities will take place: All activities are computational and take place at Wayne State University. No activity occurs at a clinical site, and no participant is contacted, recruited, observed, or enrolled at any location. Data are downloaded over the internet from public archives to WSU-managed computing resources, including the WSU High Performance Computing Grid and WSU-managed workstations. Analysis, annotation, model training, and manuscript preparation are performed on those resources. Derived annotations and software are published to public repositories (Zenodo, GitHub, Hugging Face) under open licences. Addition for this dataset: the VinDr-SpineXR files are held only on the WSU High Performance Computing Grid under the storage of the PhysioNet account holder (Gregory Schwing, PhysioNet username gschwing). Under the data use agreement they are not copied to any public repository, not shared with any person outside the agreement, and no image or derived frame from them is included in any published dataset.		
2.	Select the type of data being collected: Check all that apply:	☑ Secondary or retrospective data collection *(Existing data that was originally generated for reasons unrelated to this project)* ☐ Identifiable Protected Health Information (PHI) ☐ Prospective Data ☐ Secondary or retrospective collection of bio-specimens ☐ Prospective collection of bio-specimens	
3.	Will you require **access** to, or the **use** of any identifiable information pertaining to participants during the process of collecting data?	☐ Yes ☐ No	
4.	Will data being obtained include individually identifiable information?	☐ Yes ☐ No	If yes, list all identifiers being collected (e.g. name, date of birth, medical record number, email address, other codes; etc.): None. No individually identifiable information is obtained, used, generated, or retained. The depositors de-identified the images before release (protected health information removed from the DICOM tags; patient information burned into the pixel data removed after manual inspection of every scan; de-identified

5.	Could the identities of participants be known to, or be readily ascertained by the investigators?	☐ Yes ☑ No
6.	Select the source of data collection: Check all that apply	☐ Medical Record Review: Complete #6a. ☐ Survey ☐ Interview ☐ Bio-bank ☑ Data Repository ☑ Other Describe: PhysioNet (physionet.org), a research data repository operated by the MIT Laboratory for Computational Physiology, serving the dataset VinDr-SpineXR v1.0.0 (Pham HH, Nguyen Trung H, Nguyen HQ, 2021; doi 10.13026/q45h-5h59) under the PhysioNet Restricted Health Data License 1.5.0 to registered users who sign the PhysioNet Restricted Health Data Use Agreement 1.5.0. This is PhysioNet's restricted-access tier (registered account plus the agreement); no credentialing and no CITI training is required for it. This differs from the sources covered by 2026-269, described there as public archives releasing de-identified imaging under open licences without application or data use agreement.
6a.	Indicate the institution(s) (Covered Entity) you will be reviewing and collecting medical record data from: Check all that apply:	☐ Detroit Medical Center Facility ☐ Karmanos Cancer Institute ☐ J.D. Dingell Veterans Administration Medical Center ☐ Other Describe: N/A. No medical records at any covered entity are accessed. All imaging is obtained from the repository, which received it already de-identified from the depositors (Hospital 108 and Hanoi Medical University Hospital, Viet Nam, 2011-2020, under those hospitals' IRB approval with a waiver of informed consent). ☑ N/A- Project does not involve the review of or collection of data from a medical record
	Note: All applicable institutional policies must be followed at all times. When collecting medical record data outside of your normal responsibilities within the institution, you must obtain approval to access and collect medical record data.	

7.	Describe how the data will be obtained (e.g. survey, interview, observation, testing, review of existing records, etc.): By direct download from PhysioNet after registering an individual PhysioNet account (username gschwing, registered with the WSU e-mail go2432@wayne.edu) and accepting the PhysioNet Restricted Health Data Use Agreement 1.5.0 online. No credentialing was required. No data are collected from participants, and no records at any covered entity are accessed. The dataset was downloaded to the WSU High Performance Computing Grid on 7 September 2026 and verified against the depositors' published SHA-256 checksums (10,466 images, 0 mismatches). Disclosure: the download and use (2,071 of the 10,466 films, in cropped form, as pseudo-labelled training data for a subset of machine-learning models in a manuscript under preparation) took place after the 2026-269 tool was completed (28 August 2026) and after that determination issued (2 September 2026). Item 7 of the 2026-269 tool promised a separate submission before any agreement-governed dataset was used; this is that submission, made late, and the investigator asks the office to determine whether this secondary analysis is human participant research. If the office determines that it is, the models trained on it will be withdrawn and retrained without it. The five criteria of item 7 of the 2026-269 tool, applied to this dataset: (1) publicly available and de-identified by the source before release -- MET (listed on PhysioNet since 24 August 2021 with a DOI; de-identification as described above); (2) downloadable without application, permission, eligibility review or data use agreement -- NOT MET (a data use agreement is accepted online; no eligibility review); (3) not designated restricted-use by the source -- NOT MET by name (the source calls the licence 'Restricted Health Data License'; the restriction is the agreement's terms: research use only, no re-identification, no sharing of access, report any identifying information found); (4) no key, code or relationship enabling re-identification -- MET; (5) no re-identification attempt and no linkage -- MET (the agreement also forbids it). The agreement's obligations are honoured: files reside only in the account holder's WSU storage; no image, crop or derived frame is redistributed; the project's code is released under an open licence so the

Section C: Description of the Project:

Provide a description of the project with enough detail for the determination.

8.	Describe the purpose of this project: The Open Spine Consortium develops open, reproducible methods for analysing spinal and pelvic anatomy on computed tomography and radiographs. Its projects share one design: existing, publicly released, de-identified imaging is re-analysed to produce annotations, measurements, software, and machine-learning models, which are then released openly for others to use and check. The clinical problem motivating this work is that vertebral level is conventionally established by counting down from C2. An abdominal CT or a limited intraoperative image does not contain C2, so a transitional vertebra, a rib borne by a lumbar vertebra, and a stump rib produce overlapping appearances, and the correct label depends on a count the field of view cannot support. Mislabelling a level has direct consequences for surgical planning. The purpose of THIS determination is narrower than 2026-269. It asks whether the secondary analysis of one additional de-identified dataset, VinDr-SpineXR, which is governed by a data use agreement and therefore falls outside the scope that 2026-269 defined for itself, constitutes human participant research. The dataset's role is as additional training images of the lumbar and thoracic spine on plain radiographs for a model that names vertebral levels on intraoperative C-arm images.

9.	Describe the objectives or aims for the project: Across current and planned Open Spine Consortium projects: 1. Produce and openly release expert-reviewed anatomical annotations of existing public imaging datasets (vertebrae by level, sacrum, pelvis, femurs, ribs by level, and surgical instrumentation where present). 2. Characterise anatomical variation in these cohorts, particularly lumbosacral transitional anatomy and its effect on vertebral level assignment. 3. Develop, train, and evaluate segmentation and measurement algorithms on those annotations, and release the software and trained models under open licences. 4. Compute spinopelvic parameters reproducibly from segmentations, and evaluate agreement against established methods. 5. Report methods and results in peer-reviewed publications, with the underlying annotations and code archived publicly. No objective requires the identity of any individual, and none is served by knowing it. For this dataset only objective 3 is engaged: VinDr-SpineXR images serve as training data. No annotation of VinDr-SpineXR is released (the vertebral levels assigned to its images were generated automatically and are not distributed), and no VinDr-SpineXR image appears in any released dataset, figure or accuracy table.
10.	Describe how the results will be used/applied: Results are used to advance scientific and clinical understanding of spinal anatomy and its measurement, and are disseminated as peer-reviewed journal articles and conference papers; openly licensed datasets of derived annotations archived with a DOI; open-source software and trained models; and educational and reference material for clinicians and researchers. Findings are not applied to the care of any individual whose images appear in these datasets. No individual is contacted, and no result is returned to any individual, because no individual can be identified from the data. Software released by the consortium is provided for research use only and is not offered for clinical diagnosis or treatment; should FDA clearance later be sought for any device, a separate determination would be obtained beforehand. Addition for this dataset: the PhysioNet licence requires the licensee to contribute code associated with publications arising from the data to an open repository; the project's code is released under an open licence, which satisfies this. Trained model weights that learned from VinDr-SpineXR images do not contain, and cannot reproduce, any source image.

11.	Describe the participants for the project: There are no participants in this project as the term is used in 45 CFR 46. The project analyses imaging studies that were acquired previously, for clinical or research purposes unrelated to this work, and that were de-identified and released by their source before the investigators obtained them. The investigators have no contact of any kind with the individuals whose images these were, have no means of identifying them, and receive no key or code that would permit re-identification. For this dataset: VinDr-SpineXR comprises 10,466 spine radiographs from 5,000 studies (4,000 training studies with 8,389 images; 1,000 test studies with 2,077 images), collected retrospectively by the depositors at Hospital 108 and Hanoi Medical University Hospital over 2011-2020 under those hospitals' IRB approval with a waiver of informed consent, de-identified, and deposited on PhysioNet in 2021. The per-image information available to the investigators is the image itself, the depositors' study and image identifiers, and their lesion annotations. Of the 10,466 images, 2,071 were used in this project.

Section D: Determination Tool:

In order for a project to meet the regulatory definition of human participant (subject) research under HHS regulation 45 CFR 46.102 and FDA regulation 21 CFR 50.3, the project must meet the definition of a human participant(subject) and the definition of research. The following questions will help to determine if your project meets these definitions and requires IRB review.

Step 1: Does the Project Meet the Regulatory Definition of Research Under HHS Regulations (45 CFR 46)?

Research means a systematic investigation, including research development, testing, and evaluation, designed to develop or contribute to generalizable knowledge.

Select all that apply:

1	1.1	☑	Information is expected to expand the knowledge of a scientific discipline or other scholarly field or study and yield one or both of the following:	
			☑	Results are intended to be applicable to a larger population beyond the site of data collection or the specific participants/subjects studied.
			☑	Results are intended to be used to develop, test, or support theories, principles, and statements of relationships or to inform policy beyond the study.
	1.2	☑	The information is collected to share with others in a discipline and is created to make a broad statement (conclusion) about a group of people, procedures, programs, etc.	

Step 2: Does the Project Involve Human Participants/Subjects Under HHS Regulations (45 CFR 46)?

Human subject means a living individual about whom an investigator (whether professional or student) conducting research:

- Obtains information or bio-specimens through intervention or interaction with the individual, and uses, studies, or analyzes the information or bio-specimens; or
- Obtains, uses, studies, analyzes, or generates identifiable private information or identifiable bio-specimens

2.	2.1	Does the research involve collecting data through *intervention or **interaction with the individuals?	☐ Yes: ☐ No:
		Definitions (45 CFR 46.102): ***Intervention includes:*** • Both physical procedures by which information or bio-specimens are gathered and manipulations of the participant or the participant's environment that are performed for research purposes. ***Interactions involve:*** • Communication or interpersonal contact between the investigator and the participant. Interactions can be in-person or virtual. Examples include interactions via email or social media, or by completing a survey, questionnaire, interview or participation in a focus group.	
	2.2	Will you obtain, use, study, analyze, or generate *identifiable private information or **identifiable bio-specimens from individuals?	☐ Yes: ☐ No:
		Definitions (45 CFR 46.102): ***Identifiable private information:*** information for which the identity of the participant may be readily ascertained by the investigator or associated with the information. **Private information includes:**	

			• Information which has been provided for specific purposes by an individual and which the individual can reasonably expect will not be made public (e.g., a medical record, emails, certain listserv communications, class papers and exams, etc.) • Identifiable private information is private information for which the identity of the subject is or may readily be ascertained by the investigator or associated with the information. ****Identifiable bio-specimen** is a bio-specimen for which the identity of the subject is or may be readily ascertained by the investigator or associated with the bio-specimen.

Step 1 & Step 2 Human Participant Research Determination Decision Tool:

Step 1: Research Determination:

- If you checked any boxes in question 1.1 or 1.2, the project is considered research per HHS (45 CFR 46.102(l))

Step 2: Human Participant Determination:

- If you answered yes to question 2.1 or 2.2, the project involves human participants per HHS (45 CFR 46.102 (E))

Human Participant Research Determination:

The project is HPR and IRB Review is required if:	Step 1 finds that your project involves research **and** step 2 finds that your project involves human participants**.** • **Submission instructions and guidance are available on the IRB website:** www.research.wayne.edu/irb. • Email your questions to irbquestions@wayne.edu
The project is not HPR and does not require IRB review if either of the following applies:	• If step 1 finds that your project does not involve research and/or • If step 2 finds that your project does not involve human participants
Step 3: Proceed to Step 3 to determine if the project is subject to Federal Drug Administration (FDA) regulations.	

Step 3: Does the activity require IRB review under the FDA Regulations (21 CFR 50)?

Select all that apply

3.	3.1	☐	Activity is conducted in the United States and involves the **use of a drug** in one or more *human subjects (as recipients of a **test article or as controls, patient or healthy 21 CFR 50.3) but is **not** the use of an approved drug in the course of medical practice
	3.2	☐	Activity is conducted in the United States and evaluates the safety or effectiveness of a **device** in one or more *human subjects.
	3.3	☐	Data regarding *human subjects (including controls) will be submitted to or held for inspection by FDA as part of an application for a research or marketing permit.

	3.4	☐	Data regarding the use of a device on human specimens (including de-identified/anonymous) specimens will be submitted or held for inspection by the FDA as part of an application for a research or marketing permit

Guidance- Definition (21 CFR 50.3)

***Human Subject:** an individual who is or becomes a participant in research, either as a recipient of the test article or as a control. A subject may be either a healthy human or a patient.

****Test article:** any drug (including a biological product for human use), medical device for human use, human food additive, color additive, electronic product, or any other article subject to regulation under the Food, Drug, and Cosmetic Act.

FDA Human Subject Research Determination Instructions:

If any of the boxes in #3 are checked, the activity is human research per FDA regulations and subject to IRB review.

Note: If the activity is determined to be human research subject to FDA regulation, this does not necessarily mean it is also subject to HHS regulation, and vice versa. Separate determinations should be made.

If assistance is needed, or if written documentation from the IRB office is required:

Complete the **entire** form and email it to IRBQuestions@wayne.edu. Include any relevant supporting documents (e.g., grant, protocol, data collection tools). Please do not submit handwritten documents to the IRB office.

WSU IRB Determination:

(To be completed by IRB Administration)

☑	**Not Human Participant Research - IRB review is not required**	
	Type of activity that does not require IRB review:	
	☐	Case Report **Note:** IRB approval is required if the case report involves more than three cases
	☐	Course Related Activities **Note:** IRB Approval is required if a student is involved in an activity designed to teach research methodologies and the instructor or student wishes to conduct further investigation and analyses in order to contribute to scholarly knowledge.
	☐	Decedents: Research limited to death records, autopsy materials or cadaver specimens. **Note:** IRB approval is required if decedent information contains psychotherapy notes, or information related to HIV, mental health, genetic testing or drug or alcohol abuse
	☐	Journalism/Documentary Activities **Note:** IRB approval may be required when journalists conduct activities normally considered scientific research intended to develop generalizable knowledge (e.g. systematic research, surveys, and/or interviews that are intended to test theories or develop models).
	☐	Oral History **Note:** IRB approval is required when the activities are intended to develop generalizable conclusions (e.g., that serve as data collection intended to test economic, sociological, or anthropological models/theories)
	☐	Program Evaluation/Quality Improvement/Quality Assurance Activities **Note:** Investigators conducting QI/QA projects should ensure that they have received approval from any applicable committees within their department or the site in which the activity will occur
	☐	Public Use Datasets **Note:** IRB approval is required for the use of restricted use data, if a proposal is required to obtain the dataset, or if a data use agreement is involved.
	☑	De-Identified Private Information and/or Human Biological Specimens **Note:** IRB approval is required if the information being collected could enable the investigator to identify or readily ascertain the identity of the individual whom the private information or specimens belongs to.
	☐	Public Health Surveillance **Note:** Activities are limited to those necessary to allow a public health authority to identify, monitor, assess, or investigate conditions of public health importance
	☐	Other Project type not considered to be HPR Describe:

☐	**Project is Human Participant Research- IRB Review Required.**	
	Type of IRB required:	
	☐	Exempt IRB review is required (minimal risk research): Rationale:
	☐	Expedited IRB review is required (minimal risk research): Rationale:
	☐	Full Board IRB review is required (greater than minimal risk research): Rationale:

Reviewer Comments:	Research: The project meets the regulatory definition of research as it is likely to contribute to generalizable knowledge (45 CFR 46.102(l)). Human Subjects/participants: This project does not meet the regulatory definition of human subjects/participants (46 CFR 46.102(e)) because investigators will be analyzing de-identified data in a manner that will make it impossible to ascertain the identity of the individual's data being analyzed for research.
Authorized IRB Reviewer Signature:	Helen Craigie Digitally signed by Helen Craigie Date: 2026.09.29 14:53:13 -04'00'

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